ALKINLAR

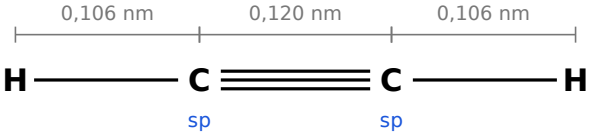
Atsetilen uglevodorodlari · umumiy formula C_nH_{2n-2}

Student book – nazariy qo‘llanma

Molekulasida uglerod atomlari orasida **bitta uchbog‘** (C≡C) bo‘lgan to‘yinmagan uglevodorodlar **alkinlar** deyiladi. Qatorning birinchi vakili atsetilen bo‘lgani uchun ular **atsetilen uglevodorodlari** deb ham ataladi. Umumiy formulasi $\mathbf{C_nH_{2n-2}}$ (n ≥ 2): alkin molekulasida mos alkendagidan 2 ta, alkandagidan 4 ta vodorod atomi kam. Alkinlar **alkadiyenlar** bilan sinflararo izomer.

Tuzilishi

- Uchbog‘dagi ikkala uglerod atomi **sp-gibridlangan**. Ikkita sp-gibrid orbital bir to‘g‘ri chiziqda yotadi: valent burchak **180°**, molekulaning bu qismi **chiziqli** tuzilishga ega.
- Uchbog‘ **1 ta σ-bog‘** va **2 ta π-bog‘**dan iborat. Har bir uglerodda gibridlanishda qatnashmagan ikkitadan p-orbital qoladi; ular yon tomonlari bilan qoplanib, o‘zaro perpendikulyar tekisliklarda joylashgan ikkita π-bog‘ni hosil qiladi.
- C≡C bog‘ C=C va C–C bog‘larga qaraganda **qisqa va mustahkam**. Shunga qaramay, π-bog‘lar hisobiga alkinlar **birikish** reaksiyalariga oson kirishadi.
- sp-gibrid orbitalda s-orbital ulushi katta (50 %), shuning uchun sp-gibridlangan uglerod sp^2- va sp^3-gibridlangan ugleroddan **elektromanfiyroq**. Natijada ≡C–H bog‘ kuchliroq qutblangan: uchbog‘i zanjir oxirida joylashgan alkinlarning vodorodi **kuchsiz kislotali** xossa namoyon qiladi va metall atomlariga almasha oladi.



H–C≡C burchagi 180°: molekula chiziqli

Atsetilen molekulasining tuzilishi

Bog‘larning taqqoslanishi

Bog‘	Misol	Gibridlanish	Burchak	Uzunligi, nm	Energiyasi, kJ/mol
C–C	etan	sp^3	109,5°	0,154	348
C=C	etilen	sp^2	120°	0,134	614
C≡C	atsetilen	sp	180°	0,120	839

Umumiy formulalar (C_nH_{2n-2})

Ko‘rsatkich	Ifoda	Misol: propin C_3H_4
Molyar massa	$M_r = 14n - 2$	40
Vodorod bo‘yicha zichlik	$D(H_2) = 7n - 1$	20
σ-bog‘lar soni	3n – 3	6
π-bog‘lar soni	2	2
C–H bog‘lari (kam qutbli)	2n – 2	4
C–C bog‘lari (σ, qutbsiz)	n – 1	2
sp / sp^3 uglerod atomlari	2 / n – 2	2 / 1
Jami atomlar soni	3n – 2	7
Yonishi	$C_nH_{2n-2} + (1{,}5n - 0{,}5)O_2 \rightarrow nCO_2 + (n-1)H_2O$	$C_3H_4 + 4O_2 \rightarrow 3CO_2 + 2H_2O$

Gomologik qator

Formula	Nomi	Formula	Nomi
C_2H_2	etin (atsetilen)	C_7H_{12}	geptin
C_3H_4	propin	C_8H_{14}	oktin
C_4H_6	butin	C_9H_{16}	nonin
C_5H_8	pentin	$C_{10}H_{18}$	dekin
C_6H_{10}	geksin	C_nH_{2n-2}	...in

Nomi mos alkan nomidagi «**-an**» qo‘shimchasini «**-in**» ga almashtirib hosil qilinadi.

1. Nomlanishi

Alkinlar uch xil nomlanadi: **sistematik (IUPAC)**, **ratsional** va **tarixiy (trivial)** nomenklatura bo‘yicha.

A) Sistematik nomenklatura (alkan → alkin)

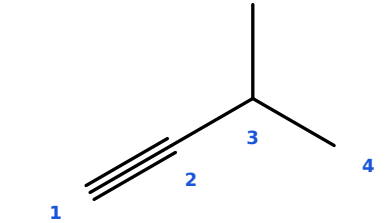
- **Uchbog‘ saqlagan eng uzun zanjir** tanlanadi. Bu zanjir molekuladagi eng uzun zanjir bo‘lmasligi ham mumkin, lekin uchbog‘ albatta unga kirishi kerak.
- Zanjir **uchbog‘ yaqin** tomondan raqamlanadi. Uchbog‘ zanjir o‘rtasida bo‘lsa, radikallar raqami eng kichik bo‘ladigan tomondan raqamlanadi.
- Radikallar **alifbo tartibida**, o‘rni (raqami) ko‘rsatilgan holda aytiladi; di-, tri-, tetra- qo‘shimchalari hisobga olinmaydi.
- Oxirida asosiy zanjirga mos alkan nomidagi «-an» o‘rniga **«-in»** qo‘shiladi va uchbog‘ o‘rni – uchbog‘dagi birinchi uglerod raqami yoziladi: butin-1, butin-2.

Eslatma: etin va propin nomida uchbog‘ o‘rni yozilmaydi, chunki boshqa variant yo‘q. Raqamlashda avval **uchbog‘**, so‘ngra radikallar hisobga olinadi.

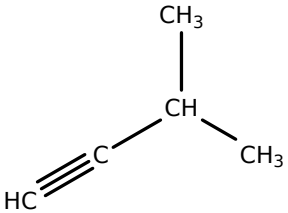
Raqamlashga misollar (skelet va yoyilgan formula)

a) 3-metilbutin-1

skelet formula

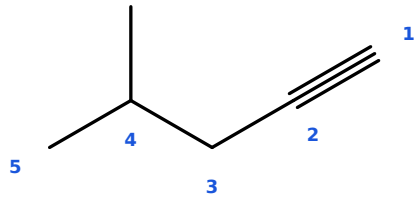


yoyilgan (to‘liq) formula

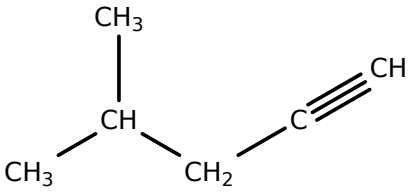


b) 4-metilpentin-1 (uchbog‘ o‘ng tomonda – raqamlash o‘ngdan boshlanadi)

skelet formula

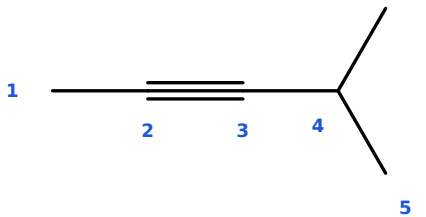


yoyilgan (to‘liq) formula

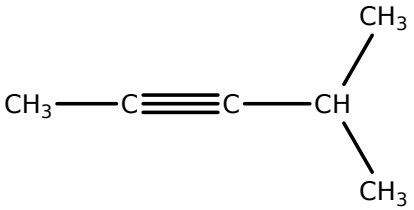


c) 4-metilpentin-2

skelet formula

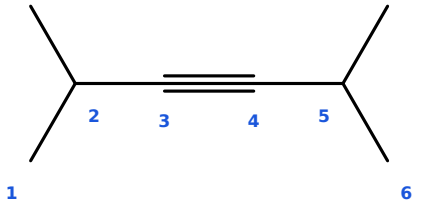


yoyilgan (to‘liq) formula

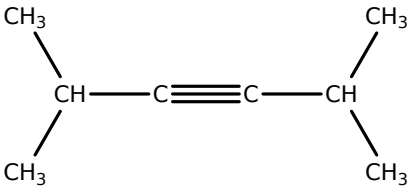


d) 2,5-dimetilgeksin-3 (uchbog‘ o‘rtada, molekula simmetrik)

skelet formula

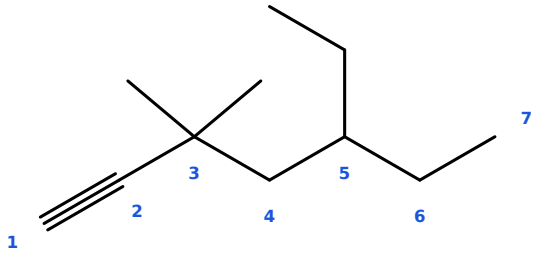


yoyilgan (to‘liq) formula

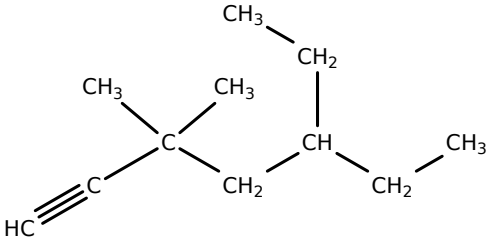


e) 5-etil-3,3-dimetilgeptin-1

skelet formula



yoyilgan (to‘liq) formula



B) Ratsional nomenklatura

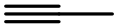
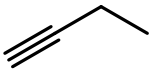
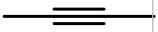
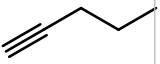
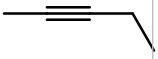
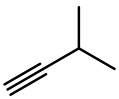
Eng oddiy vakil – **atsetilen HC≡CH** asos qilib olinadi va uning vodorod atomlari o‘rnini radikallar egallagan deb qaraladi. Nomi radikallar nomi va «atsetilen» so‘zidan tuziladi: $HC{\equiv}C{-}CH_3$ – metilatsetilen, $CH_3{-}C{\equiv}C{-}CH_3$ – dimetilatsetilen, $CH_3{-}C{\equiv}C{-}C_2H_5$ – metiletilatsetilen.

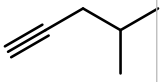
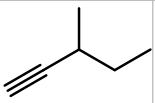
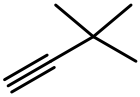
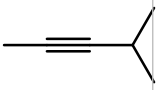
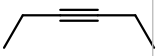
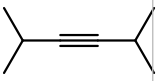
C) Tarixiy nomlar va radikallar

Ayrim alkinlar tarixiy nomlari bilan ham ataladi: $HC{\equiv}CH$ – **atsetilen**, $HC{\equiv}C{-}CH_3$ – **allilen**, $CH_3{-}C{\equiv}C{-}CH_3$ – **krotonilen**, $HC{\equiv}C{-}CH{=}CH_2$ – **vinilatsetilen**.

Radikal	Formula	Qaysi alkindan
etinil	$HC{\equiv}C{-}$	atsetilen
propargil (propin-2-il)	$HC{\equiv}C{-}CH_2{-}$	propin
propinil (propin-1-il)	$CH_3{-}C{\equiv}C{-}$	propin

Nomlash misollari (to‘plam jadval)

Oddiy formula	Skelet formula	Sistematik	Ratsional / tarixiy
$HC{\equiv}CH$		etin	atsetilen
$HC{\equiv}C{-}CH_3$		propin	metilatsetilen (allilen)
$HC{\equiv}C{-}CH_2{-}CH_3$		butin-1	etilatsetilen
$CH_3{-}C{\equiv}C{-}CH_3$		butin-2	dimetilatsetilen (krotonilen)
$HC{\equiv}C{-}(CH_2)_2{-}CH_3$		pentin-1	propilatsetilen
$CH_3{-}C{\equiv}C{-}CH_2{-}CH_3$		pentin-2	metiletilatsetilen
$HC{\equiv}C{-}CH(CH_3){-}CH_3$		3-metilbutin-1	izopropilatsetilen

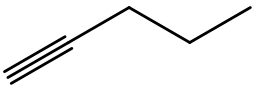
Oddiy formula	Skelet formula	Sistematik	Ratsional / tarixiy
$HC\equiv C–CH_2–CH(CH_3)–CH_3$		4-metilpentin-1	izobutilatsetilen
$HC\equiv C–CH(CH_3)–CH_2–CH_3$		3-metilpentin-1	ikkilamchi butilatsetilen
$HC\equiv C–C(CH_3)_2–CH_3$		3,3-dimetilbutin-1	uchlamchi butilatsetilen
$CH_3–C\equiv C–CH(CH_3)–CH_3$		4-metilpentin-2	metilizopropilatsetilen
$CH_3–CH_2–C\equiv C–CH_2–CH_3$		geksin-3	dietilatsetilen
$(CH_3)_2CH–C\equiv C–CH(CH_3)_2$		2,5-dimetilgeksin-3	diizopropilatsetilen

2. Izomeriyasi

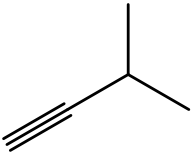
Alkinlarda uglerod **zanjiri**, uchbog‘ **holati** va **sinflararo** izomeriya uchraydi. Fazoviy (sis-trans) izomeriya alkinlarda bo‘lmaydi.

1) Uglerod zanjiri izomeriyasi

Izomerlar uglerod zanjirining tuzilishi (tarmoqlanishi) bilan farq qiladi. Bu izomeriya C_5H_8 dan boshlanadi:



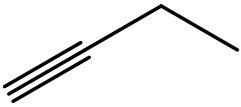
pentin-1



3-metilbutin-1

2) Uchbog‘ holati izomeriyasi

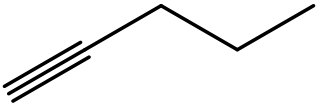
Izomerlar uchbog‘ning zanjirdagi o‘rni bilan farq qiladi. Bu izomeriya C_4H_6 dan boshlanadi:



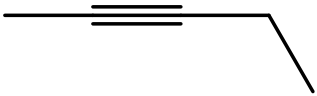
butin-1



butin-2



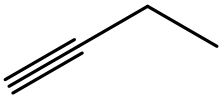
pentin-1



pentin-2

3) Sinflararo izomeriya

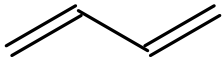
Alkinlar C_nH_{2n-2} umumiy formulali **alkadiyenlar** va **sikloalkenlar** bilan izomer. C_4H_6 tarkibli moddalarga misollar:



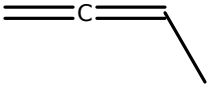
butin-1 (alkin)



butin-2 (alkin)



butadiyen-1,3 (alkadiyen)



butadiyen-1,2 (alkadiyen)

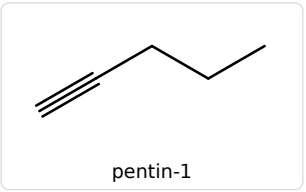


siklobuten (sikloalken)

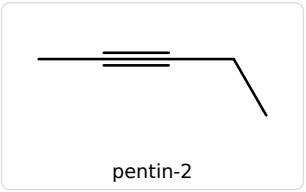
4) Fazoviy izomeriya

Uchbog‘dagi uglerod atomlarining har biri faqat **bitta** o‘rinbosar bilan bog‘langan va molekulaning bu qismi chiziqli. Shuning uchun alkinlarda **sis-trans izomeriya bo‘lmaydi** – bu ularning alkenlardan muhim farqi.

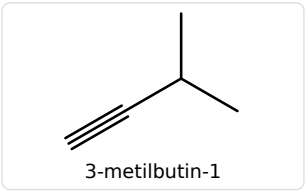
C_5H_8 tarkibli alkinlar (3 ta)



pentin-1

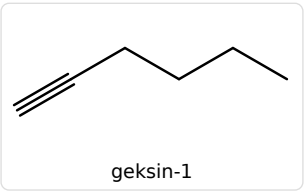


pentin-2

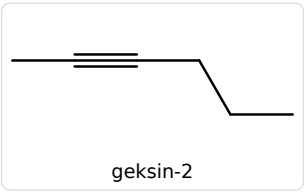


3-metilbutin-1

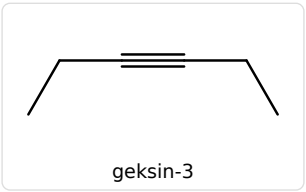
C_6H_{10} tarkibli alkinlar (7 ta)



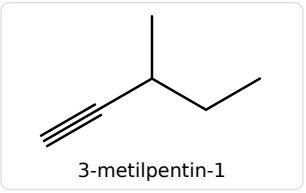
geksin-1



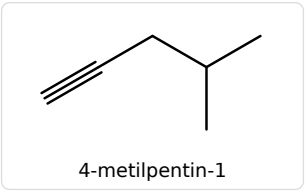
geksin-2



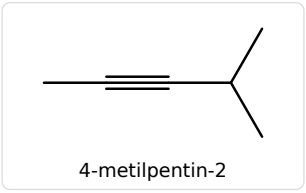
geksin-3



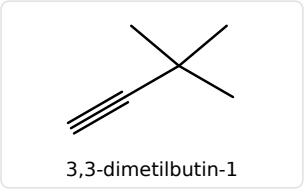
3-metilpentin-1



4-metilpentin-1



4-metilpentin-2



3,3-dimetilbutin-1

C_5H_8 va C_6H_{10} tarkibli alkadiyen va sikloalkenlar ham mavjud – ular yuqoridagi alkinlar bilan sinflararo izomer.

3. Fizik xossalari

- Xona haroratida atsetilen, propin va butin-1 – **gaz**; butin-2 dan C_{15} gacha – **suyuqlik**; C_{16} dan yuqorisi – **qattiq** moddalar.
- M_r ortishi bilan qaynash (t_q) va suyuqlanish (t_s) haroratlari hamda zichlik ortadi.
- Uchbog‘ning holati t_q ga kuchli ta’sir qiladi: uchbog‘ zanjir o‘rtasiga yaqinlashgani sari t_q ortadi (butin-1 – 8 °C, butin-2 – 27 °C).
- Alkinlarning t_q mos alkan va alkenlarnikidan biroz yuqori (etan –89 °C, etilen –104 °C, atsetilen –84 °C).

- Suvda kam eriydi, organik erituvchilarda yaxshi eriydi. Atsetilen suvda etilenga qaraganda yaxshiroq, atsetonda esa juda yaxshi eriydi.
- Atsetilen – rangsiz, deyarli hidsiz gaz, havodan biroz yengil (M = 26 g/mol). Kalsiy karbiddan olingan texnik atsetilen aralashmalar (PH_3, H_2S) hisobiga yoqimsiz hidli bo‘ladi.

Modda	Formula	Holati*	t_q, °C
atsetilen	C_2H_2	gaz	–84
propin	C_3H_4	gaz	–23
butin-1	C_4H_6	gaz	8
butin-2	C_4H_6	suyuq	27
pentin-1	C_5H_8	suyuq	40
pentin-2	C_5H_8	suyuq	56
geksin-1	C_6H_{10}	suyuq	71

* xona haroratida (20–25 °C)

4. Olinishi

1) Kalsiy karbidga suv ta’sir ettirish (karbid usuli)

$$CaC_2 + 2H_2O \longrightarrow Ca(OH)_2 + C_2H_2\uparrow$$

Kalsiy karbid so‘ndirilmagan ohak bilan koksni elektr pechida (≈ 2000 °C) qizdirib olinadi: $CaO + 3C \rightarrow CaC_2 + CO$. Atsetilen laboratoriyada ham, sanoatda ham shu usulda olinadi. Magniy karbiddan esa propin hosil bo‘ladi:

$$Mg_2C_3 + 4H_2O \longrightarrow 2Mg(OH)_2 + CH_3{-}C{\equiv}CH$$

2) Metanni piroliz qilish (sanoat usuli)

$$2CH_4 \xrightarrow{1500\ °C} C_2H_2 + 3H_2$$

Reaksiya juda qisqa vaqt davomida o‘tkaziladi va hosil bo‘lgan gazlar tez sovitiladi, aks holda atsetilen uglerod va vodorodga parchalanib ketadi.

3) Uglerod va vodoroddan sintez (M. Bertlo, 1862-yil)

$$2C + H_2 \xrightarrow{\text{elektr yoyi}} C_2H_2$$

Elektr (volt) yoyi orasidan vodorod o‘tkazilganda atsetilen hosil bo‘ladi. Atsetilen oddiy moddalardan birinchi marta shu usulda sintez qilingan.